

The non-equilibrium thermodynamics of quantal response:

criticality escape, a supercritical wedge, and the decoupling of response asymmetry from dissipation

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Abstract

Logit revision dynamics on a finite game define a Markov jump process on the joint profile space whose stationary state is a genuine thermodynamic object: a Gibbs equilibrium exactly when the game is potential, and a dissipative non-equilibrium steady state otherwise. We sweep the two-parameter plane spanned by the harmonic fraction α of the game’s Hodge decomposition and the logit precision λ , reading four calibrated meters at every point — spectral criticality $\rho(SB)$, reciprocity defect $\mathcal{R}$, exact entropy production σ_{EP} , and its trajectory estimators — and report three facts we did not predict. (i) *Criticality escape*: on potential games $\rho(SB)$ is non-monotone in λ — equilibrium concentration kills the choice covariance faster than λ amplifies it — while harmonic games cross $\rho = 1$ and stay there, opening a *supercritical wedge* whose frontier descends in α (median onset $\alpha = 0.5$). (ii) *Decoupling*: dissipation and response asymmetry co-move almost perfectly along α ($\rho_S = 0.993$ marginally) yet, conditional on α , their correlation degrades and *reverses* to -0.355 in the near-harmonic regime; a mechanism experiment shows this is structural — the equilibrium susceptibility is a local derivative, the entropy production a global flux functional, and no renormalisation of the former recovers the latter’s ordering. A repair hypothesis we proposed is refuted by its own preregistered test. (iii) *Estimability*: irreversibility is recoverable from trajectories alone — a k -th order Markov KLD estimator tracks the exact meter with rank correlation 1.0 across the family, and a debiased finite-time thermodynamic-uncertainty bound certifies dissipation from two cumulants of one current, saturating (ratio $\rightarrow 1$) precisely in the near-equilibrium regime. Every number regenerates from fixed seeds; every claim carries its confidence tier and its falsifier.

1 Setup: the joint-profile chain and its four meters

A finite game with players $i = 1..N$ and payoff tensors u_i defines *Glauber-logit dynamics*: revision opportunities arrive at unit rate per player, and the revising player redraws an action from $\text{softmax}(\lambda u_i(\cdot, a_{-i}))$. This is a continuous-time Markov jump process on the joint profile space $\prod_i A_i$. Two exact facts anchor everything below (both are permanent regression tests in the library, verified to 10^{-10} – 10^{-16} ; tiers per the claims ledger).

*With agent-assisted implementation. Code, gate records, and every number’s regeneration path: <https://github.com/SharathSPHD/sage>.

Potential games are thermal equilibrium. If the game admits an exact potential Φ , the dynamics are reversible with Gibbs stationary law $\pi \propto e^{\lambda\Phi}$ [1]: detailed balance holds, the stationary probability current $J^*(a, a') = \pi(a)w(a \rightarrow a') - \pi(a')w(a' \rightarrow a)$ vanishes edge-by-edge, and the Schnakenberg entropy production rate [4]

$$\sigma_{\text{EP}} = \frac{1}{2} \sum_{a, a'} [\pi(a)w(a \rightarrow a') - \pi(a')w(a' \rightarrow a)] \log \frac{\pi(a)w(a \rightarrow a')}{\pi(a')w(a' \rightarrow a)} \geq 0 \quad (1)$$

is exactly zero.

Harmonic content is the drive. The Hodge decomposition [2] splits any game into potential, harmonic and non-strategic components; we write $\alpha \in [0, 1]$ for the harmonic fraction of the normalised game. At $\alpha > 0$ the same dynamics settle into a non-equilibrium steady state: stationary but circulating ($J^* \neq 0$), dissipating at rate $\sigma_{\text{EP}} > 0$.

The meters. At each (α, λ) we read: (1) the *feedback gain* $\rho(SB)$ — spectral radius of the strategic feedback operator in $\chi^{\text{eq}} = (I - SB)^{-1}S$, with $\rho \geq 1$ marking a near-singular resolvent (criticality); (2) the *reciprocity defect* $\mathcal{R} = \|\chi^{\text{eq}} - \chi^{\text{eq}\top}\|_F / \|\chi^{\text{eq}} + \chi^{\text{eq}\top}\|_F$, which is zero iff the game is potential, at every λ [12]; (3) the *exact* σ_{EP} of the joint-profile chain; and (4) trajectory estimators of σ_{EP} that see only an observed path (Section 4). Families with α prescribed exactly are generated by unit-normalising the potential component of one random game and the harmonic component of another and mixing $(1 - \alpha)\hat{P} + \alpha\hat{H}$; all sweeps below use 100 games per α level at seed 20260808 (regeneration: `make reproduce`).

Positioning. The behaviour of learning dynamics between the potential and harmonic extremes is an acknowledged obstacle [10]. Our angle is complementary and, to our knowledge, unoccupied: treat the stationary state *thermodynamically* — currents, dissipation, response symmetry, uncertainty bounds — with instruments calibrated where the correct reading is provable.

2 The $\alpha \times \lambda$ phase map: criticality escape and the supercritical wedge

Naively $\rho(SB) \propto \lambda$, since $S = \lambda C$ with $C = \text{diag}(\sigma) - \sigma\sigma^\top$ the choice covariance. The sweep ($9 \times 11 \alpha$, medians over 100 games per cell, 9,900 solves) says otherwise (Figure 1).

Finding 1 (Criticality escape; F-0006). *On potential games ($\alpha = 0$) the median $\rho(SB)$ is non-monotone in λ : $0.07 \rightarrow 0.51$ (at $\lambda = 1.7$) $\rightarrow 10^{-3}$ (at $\lambda = 15$). As $\lambda \rightarrow \infty$ the principal QRE branch concentrates on a strict equilibrium, $C \rightarrow 0$ exponentially, and $S = \lambda C \rightarrow 0$ despite the λ factor: the system escapes criticality through equilibrium concentration. At $\alpha = 1$, mixed equilibria keep C bounded away from zero and ρ grows without bound (6.9 at $\lambda = 15$).*

Finding 2 (The supercritical wedge; F-0006). *Between the extremes a wedge of supercritical games opens. The median game first crosses $\rho = 1$ at $\alpha = 0.5$ (for $\lambda \geq 8.5$); a 0.2 fraction of games crosses already at $\alpha = 0.4$; the frontier $\lambda_c(\alpha)$ descends monotonically ($\lambda_c \approx 3$ by $\alpha = 0.8$). Inside the wedge the resolvent is near-singular, so response magnitudes (χ^{eq} , and hence $\mathcal{R}$ levels) are unreliable there — the meters themselves flag it; symmetry statements survive as direction-only reads.*

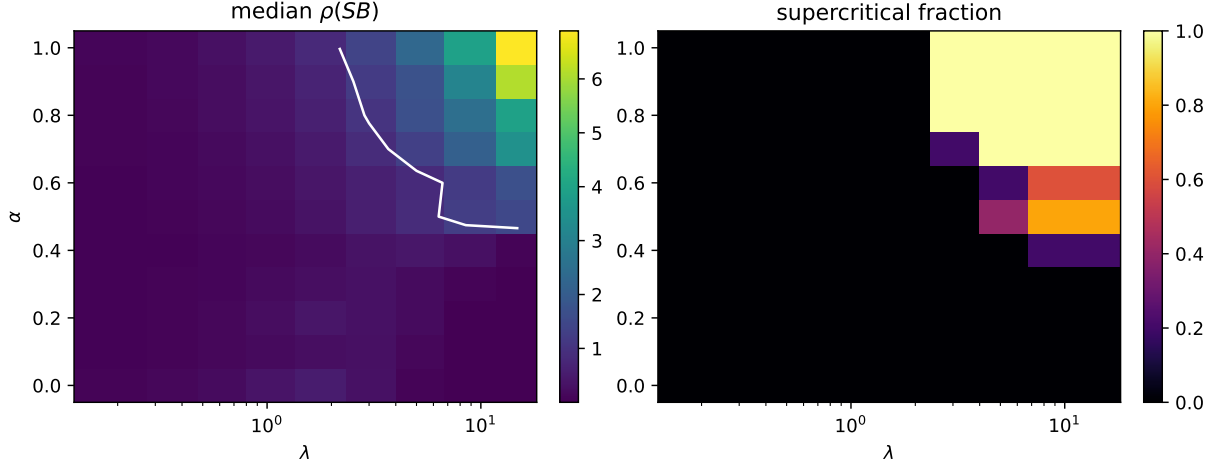


Figure 1: The $\alpha \times \lambda$ plane. Left: median $\rho(SB)$ with the $\rho = 1$ contour in white — non-monotone along the bottom edge (criticality escape), unbounded along the top. Right: fraction of games per cell with $\rho \geq 1$ — the supercritical wedge, frontier descending in α . Artifact: `phase_map_surface.json`.

Two consistency checks the map passes: the escape mechanism predicts no Hopf bifurcation of logit dynamics in full potential games ($B = B^\top$ makes the SB spectrum real; verified exactly across the λ grid), consistent with cycles requiring non-potential structure [11]; and the $\alpha = 0$ row of every dissipation meter reads zero to machine precision at all λ , as it must.

Open at this draft: whether the $\alpha = 0$ peak location ($\lambda \approx 1.7$ at payoff scale 2.0) is a pure scale artefact of the λ -payoff folding, and whether $\lambda_c(\alpha)$ is a clean curve under a finer grid.

3 Response asymmetry and dissipation are distinct observables

Both $\mathcal{R}$ and σ_{EP} vanish exactly on potential games and read loudly on harmonic ones, so a natural conjecture (C1 in the programme’s ledger) held that the four meters co-move tightly along the whole family. Marginally they do: over 1,000 games (100×10 α levels, $\lambda = 1.2$), Spearman $\rho(\sigma_{EP}, \mathcal{R}) = 0.993$.

Finding 3 (Conditional decoupling and sign reversal; F-0004). *The marginal agreement is α -confounding. Conditional on the level, $\rho(\sigma_{EP}, \mathcal{R})$ is $+0.80..+0.88$ for $\alpha \leq 0.65$, degrades through $+0.61$ ($\alpha = 0.75$) and $+0.32$ ($\alpha = 0.85$), and reverses to -0.355 at $\alpha = 0.95$: among near-pure-harmonic games, larger response asymmetry associates with less dissipation (Figure 2). This finding originated as a red-team objection to our own headline correlation; the stratified metric is now a permanent artifact field.*

Mechanism, by preregistered test (F-0007). We proposed a repair: perhaps only $\mathcal{R}$ ’s denominator $\|\chi^{eq} + \chi^{eq^\top}\|$ misbehaves at high α (H2), while the antisymmetric numerator keeps tracking dissipation (H1) — in which case a numerator-only meter would restore coupling. On the identical seeded families, paired by game: **H2 confirmed** — at $\alpha = 0.95$, $\rho(\mathcal{R}, 1/\|\chi^{eq} + \chi^{eq^\top}\|) = 0.993$, rising monotonically from ≈ 0 at low α ; $\mathcal{R}$ ’s within-level variation in the near-harmonic regime is entirely the shrinking symmetric response. **H1 refuted** — the

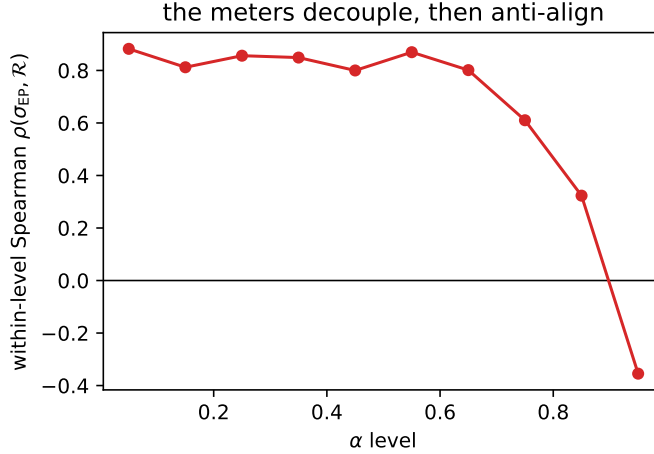


Figure 2: Within-level rank correlation between σ_{EP} and $\mathcal{R}$ along α . Strong coupling while a potential component modulates both meters; decoupling and sign reversal as it vanishes. Artifact: `chain.comovement.json`.

numerator *also* decouples: $\rho(\|\chi^{\text{eq}} - \chi^{\text{eq}\top}\|, \sigma_{\text{EP}}) = -0.37$ at $\alpha = 0.95$, down from $+0.77..0.86$ at $\alpha \leq 0.65$. No renormalisation of the response matrix recovers the dissipation ordering.

The structural reading. This negative result is the better result. χ^{eq} is a *local derivative* of the equilibrium map at the QRE point; σ_{EP} is a *global functional of the stationary flux* of the joint-profile chain. They co-vary exactly as long as a potential component modulates both, and part company when it vanishes. Consequently the instrument-scope table of [12] is not a convenience but a necessity: $\mathcal{R}$ answers “is this system potential?” (exactly, at every λ); σ_{EP} answers “how hard is it driven?”; and in the near-harmonic regime neither substitutes for the other. Conjecture C1 is thereby resolved as: co-movement for $\alpha \lesssim 0.65$, structural failure above, with the crossover’s λ - and size-dependence still open.

4 Dissipation from trajectories alone

Everything above used the generator. Data never provides it: only an observed sequence of profiles. Two estimators bridge the gap, each validated against the exact Schnakenberg meter on synthetic Glauber trajectories *before* any data contact (unit `thermo.estimators`; red-team signed off after one blocking objection, disclosed below).

Sampling identity. We simulate by uniformisation: the skeleton chain $P = I + L/\Lambda$ stepped with $\text{Exp}(\Lambda)$ holding times. Two exact facts make the skeleton the right estimation object: it shares the chain’s stationary law, and its per-step entropy production equals $\sigma_{\text{EP}}/\Lambda$; moreover the $(k+1)$ -block KLD of a stationary Markov chain equals k times the per-step entropy production, so the estimator below has a sharp target at every k .

KLD k -th order Markov [7]. The plug-in $\hat{\sigma}^{(k)} = \frac{1}{k\tau} \sum P(Y_{0:k}) \log P(Y_{0:k})/P(Y_{k:0})$ recovers the exact meter with Spearman $\rho = 1.0$ and $\sim 1\%$ per-level error across the ten- α family (Figure 3, left), and reads $< 5 \times 10^{-3}$ on exact potential games (pure plug-in bias; truth is 0).

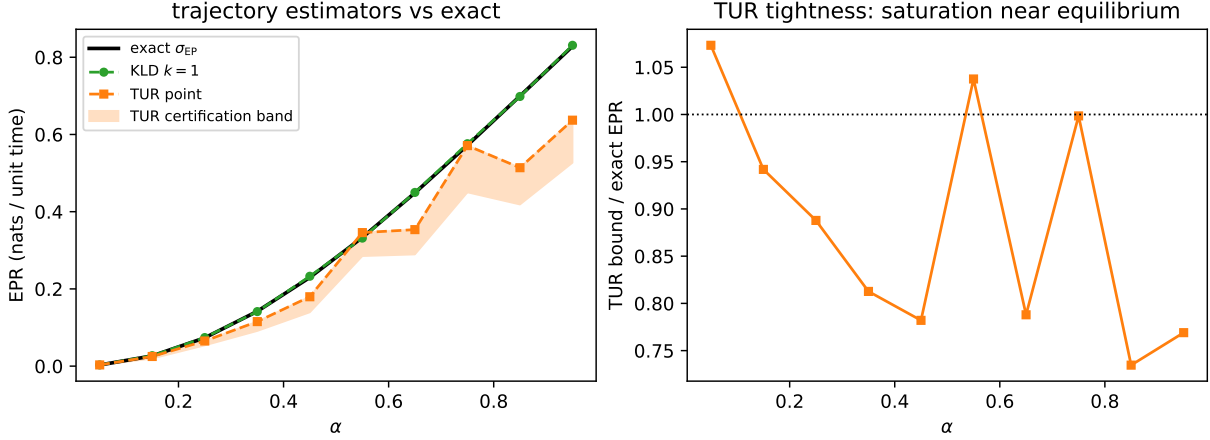


Figure 3: Left: exact EPR vs. its trajectory estimators along α ($\lambda = 1.5$; KLD from $8 \times 60k$ stationary steps, TUR from 128 fixed-horizon windows). Right: TUR tightness — the bound saturates near equilibrium and degrades gracefully (and conservatively) as the drive grows. Artifact: `estimator_alpha_sweep.json`.

It is data-hungry: the identity is a population statement, and with $n \not\gg n_{\text{states}}^{k+1}$ the estimator *underestimates* ($k = 4$ reads $\sim 17\%$ low at our validation sample size; the failure mode is itself a regression test).

TUR lower bound [5, 6]. From the first two cumulants of a time-integrated current J_T : $\sigma_{EP} \geq 2\langle J_T \rangle^2 / (\text{Var}(J_T) T)$. A certified bound rather than a point estimate — a poor current choice loosens but never breaks it — which makes it the designated headline number for partially observed data [8]. Three methodological facts were learned adversarially and are now tests: (i) the bound is a *fixed-horizon* statement — fixing the jump count instead suppresses the Poisson timing fluctuation and inflated the “bound” $1.5\times$ above the true EPR before we caught it; (ii) at M windows the sample version carries $+\text{Var}/M$ and Jensen-on- $1/\widehat{\text{Var}}$ biases, both now corrected; (iii) even debiased, the *point* estimate legitimately straddles the truth near equilibrium, where the TUR saturates — it exceeded the exact EPR at 2 of 10 sweep levels (max ratio 1.073). The certified statement is the lower bootstrap quantile, which stayed below the exact EPR at every level. The initial artifact language called the point estimate a certified bound; the red-team blocked the gate until the distinction was made structural (a separate `tur_epr_bound.ci` API) — we record this because the correction is part of the result.

Tightness is a diagnostic. The bound/exact ratio is ≈ 0.97 at $\alpha = 0.05$ (linear-response saturation) and drifts to 0.6–0.7 by $\alpha = 0.95$ (Figure 3, right): the TUR is tightest exactly where dissipation is smallest, i.e. it loses efficiency precisely where there is more to measure — the right failure direction for a conservative instrument.

5 Discussion

Three findings, one arc. The phase map says *where* the interesting physics lives: not at the calibration anchors but in the wedge between them, where strategic feedback goes supercritical

and response magnitudes stop being trustworthy. The decoupling result says *what* lives there: the response layer and the dissipation layer, interchangeable to the eye along the marginal sweep, are structurally different observables — a local derivative and a global flux — and the near-harmonic regime is where the difference becomes measurement-relevant. The estimator result says the dissipation layer is *reachable from data*: rank-exact recovery when the chain is observed, and a certified lower bound that survives partial observation, with its tightness itself carrying regime information.

What would change our minds. The wedge frontier could dissolve under payoff-scale folding (the $\alpha = 0$ peak at $\lambda \approx 1.7$ is suspiciously close to $\text{scale}/2$); the decoupling crossover at $\alpha \approx 0.7$ could move materially with λ or with action-space size, which would demote it from structural fact to small-game artefact; and the suspicious coincidence that F-0004’s decoupling onset sits *inside* the supercritical wedge deserves either a mechanism or a debunking. All three are queued as finer sweeps with committed seeds.

Toward data. The empirical programme (Stage 3 of the repository) points these meters at systems whose α nobody knows: retail pricing panels and electricity bidding. The division of labour established here is what makes that honest: the reciprocity test for potentiality (exact, λ -free as a symmetry statement), the TUR bound for dissipation (certified, partial-observation-tolerant), and neither doing the other’s job. The nearest empirical precedent is the entropy-production detection of an Edgeworth price cycle in experimental data [9]; the instruments here are built to make such readings routine, calibrated, and falsifiable.

Provenance

Every quantitative claim in this draft maps to a committed, gate-checked artifact regenerated from fixed seeds by `make reproduce`:

Claim	Artifact	
Phase surface, wedge, criticality escape	<code>phase_map_surface.json</code>	s
Marginal and within-level $\rho(\sigma_{\text{EP}}, \mathcal{R})$	<code>chain_comovement.json</code>	d
H1/H2 mechanism test	<code>decoupling_mechanism.json</code>	s
KLD/TUR ground truth (named games)	<code>estimator_ground_truth.json</code>	t.
KLD $\rho = 1.0$; TUR certification per level	<code>estimator_alpha_sweep.json</code>	t.
Zero calibration (potential games, real network)	<code>equilibrium_reads_zero.json, sioux_falls_calibration.json</code>	d

Each gate closed only after adversarial review; the objections and their dispositions (including the blocking objection to the TUR certification language, and the red-team origin of the decoupling finding itself) are in `gates/*.yaml` in the repository. Claims carry tiers in `memory/claims.md`; conjecture C1’s partial falsification is recorded there, not smoothed over.

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